

Contact-Hour Companion — Sizing a Module to Its Time

*The tool for two moments in the build: the **coarse fit-check** in §3.5 (Decompose Into Modules) — a ballpark size check from a module’s objectives and depth, before any materials exist — and the **precise time budget** in §5 (Structure the Module), once the materials are real and you can sum actual minutes. Either way it estimates how long a module’s work takes, so you can check it fits your container. Built on the Georgetown SCS Contact Hour Assessment, **re-pointed for short professional workshops** (not 15-week academic courses).*

Read this first: these are not academic courses

The original Georgetown tool sizes a **15-week graded academic course** by its *weekly out-of-class workload* — a 7–9 hr/week target full of discussion-board posts, reflection papers, and HBR readings. **Your courses are different:**

- They’re **defined by contact time**, not weekly homework load. The session length *is* the container.
- They’re **short and varied** — a 1.5-hour intro, a 3-hour workshop, a 5-week 2-hr/week series.
- They’re **hands-on and ungraded** — practice and demos, not graded discussion posts.

So you use the tool’s **mechanism** (estimate time, sum, compare to the container) and its **reading-speed benchmarks**, but you **drop the academic furniture** (discussion-post counts, reflection-paper hours, weekly-workload targets). The question is simply: **does what I planned fit the time I actually have?**

The tool works in both directions

A “container” is your course’s real schedule, measured in **total contact hours**. You don’t have to use any standard format — **design whatever shape fits your goal**. This tool just lets you reason about time both ways:

- **Time-first** — “*I have this much time. What can I realistically cover?*” (Fit the content to the container.)
- **Content-first** — “*I want to teach this. How much time will it take?*” (Let the content size the container.)

Most builds bounce between the two until content and time agree.

Example formats (these are formats I have used — yours can be anything)

Format	In-class contact	+ Out-of-class	Total	Rough # modules
Intro to a topic	1.5 hr	—	1.5 hr	1
Workshop	3 hr	—	3 hr	1
Lecture series	5 × 1 hr	—	5 hr	~5
5-week series	5 × 2 hr	optional	~10 hr	~5
8-week course	8 × 3 hr	8 × 1 hr	~32 hr	~8
12-week course	12 × 4 hr	(in the 4)	~48 hr	~12

These are **starting points, not a menu**. The module count falls out of (total hours ÷ how long each coherent chunk needs) — a 3-hour workshop is likely one module; a 12-week course, about twelve.

The formula

$$\text{Hours} = \text{Total quantity} \div \text{Average-per-hour}$$

For each reading, video, demo, or practice activity: estimate how much a student gets through in an hour (**average per hour**), how much there is (**total**), and divide.

Benchmark rates (starting estimates — tune to your audience)

Readings (from the Georgetown tool — pages/hour, slower for denser material):

Material	Pages/hour	What the student is doing
Dense / technical (textbook-style)	~20	Studying — reading to learn and retain
Article (HBR-style)	~30	Reading closely, but not to memorize
Web / lighter reading	~35	Reading — following the argument once

*Name the purpose or the number means nothing: it's the single biggest factor here, far bigger than how fast your people read. Measured adult silent reading of non-fiction averages about 238 words per minute, but measured **study** rates run from roughly 170 down to 40 wpm depending on the task (Brysaert 2019, a meta-analysis of 190 studies). At ~450 words a page, the rows above work out to about 150, 225 and 260 wpm — so “~20 pages/hour” is a studying rate and “~35” is a reading rate. They're in the right places; just be honest about which one you're asking for.*

Watch / listen (*count at real length*):

Activity	Rate
Recorded video / lecture	1 × its length (a 12-min video = 0.2 hr) + note time
Synchronous session / live demo	1:1 (1 hr = 1 hr)
Podcast / audio	1 × its length

Hands-on (the part academic tools miss — estimate per rep, then tune):

Activity	Starting estimate
Guided hands-on rep (a brew, a build, one exercise)	~15–20 min each
Practice / worksheet set	estimate by # of problems
Paired / group activity	real time it runs

Two of those rows can't be used at the §3.5 fit-check — “real time it runs” and a 1:1 live demo both need materials that don't exist until §6. That's expected: at §3.5 you're sizing, so estimate; at §6a step 6 you're budgeting, so measure.

Shared equipment changes the arithmetic. If groups share a machine, a bench or a station, elapsed time isn't the length of the task — it's **machine-minutes × number of groups ÷ number of machines**. Three pairs sharing one mill at twenty minutes each is a sixty-minute block, not a twenty-minute one. Plan a bench task for the groups who are waiting, or that time becomes dead air.

(The original academic rates — reflective response ~ 1 hr, peer responses ~ 2/hr — are kept here only for reference; most won't appear in a hands-on workshop.)

How long will this take *them*? Don't estimate — instrument.

The benchmarks above get you a starting number. They can't get you the right one, because the person estimating is the person who already knows how to do it.

You will underestimate, and your expertise is the reason. This is a measured effect, not a figure of speech: when experts, intermediates and novices were each asked to predict how long novices would take on a task, **the more expertise someone had, the worse their prediction was** — and standard debiasing instructions didn't fix it (Hinds, 1999, *Journal of Experimental Psychology: Applied*). People at an intermediate level were the most

accurate. Worse, experts also write instructions at a higher level of abstraction, so the same expertise that mis-times the task also *specifies* it in a way that slows novices down (Hinds, Patterson & Pfeffer, 2001, *Journal of Applied Psychology*).

Padding won't save you. When thesis students predicted their own submission dates, the average prediction was 34 days and the average actual was 56. Fewer than half finished even by their own **worst-case** date — and predictions made deliberately pessimistically were no more *accurate* than the plain ones (Buehler, Griffin & Ross, 1994, *Journal of Personality and Social Psychology*). Adding 50% to a number you made up gives you a bigger made-up number.

There is no multiplier to reach for. You'd want a rule like “novices take three times as long.” It doesn't exist. The measured expert-novice gap ranges from *no difference at all* to roughly fourfold depending on the domain and the task, and the famous “four times” figure comes from a study comparing **one** expert with **one** novice. Any general multiplier would be invention dressed as evidence.

So do this instead:

1. **Run the activity yourself with a clock running.** Write down the wall-clock time — don't trust the memory of it afterwards; recalled durations are themselves systematic underestimates (Roy, Christenfeld & McKenzie, 2005, *Psychological Bulletin*). That number is your **floor**, not your estimate.
2. **Put it in front of two or three people who actually resemble your learners**, and log their time from a timestamp. Don't ask them afterwards — in one study learners over-reported how long work took them 78% of the time, by a median factor of about 1.45, while 79% of them believed they tracked their time accurately (Hart, Warren & Edwards, 2025, *PLOS ONE*).
3. **Keep the ratio between your time and theirs.** That is your correction factor — for your domain, your materials, your people. It is the only one with any standing, because it's built from comparable cases rather than from a general rule.
4. **Write it down and reuse it.** By your third or fourth activity you're estimating from your own data instead of your intuition, which is the only correction this problem actually responds to.

None of this is exotic: timing real learners is a standard step in systematic instructional design (the small-group stage of formative evaluation collects completion time) — it's just the step almost everyone skips. If you genuinely can't find two representative people, use the slowest plausible rate from the table above, state the result as a range rather than a number, call it provisional, and measure it the first time you run the course.

How to use it

1. **List every piece of work** in the module (readings, watch/listen, hands-on).
2. **Set an average-per-hour** for each from the benchmarks above (adjust for your people).
3. **Enter the total quantity.**
4. **Compute hours** (total $\div$ average) and **sum.**
5. **Compare to the container** and adjust: overflow $\rightarrow$ split, cut depth (§7), or move to pre/post-class; too thin $\rightarrow$ merge or add practice.

The table

Type	Item	Average per hour	Total	Hours
Reading	...	pages/hr	pages	...
Watch/listen	...	$1 \times$ length	minutes	...
Hands-on	...	reps/hr	reps	...
			Module total	... hrs

Worked example — Coffee “Setup & First Brew” (90-min workshop session)

Type	Item	Avg/hr	Total	Hours
Watch (pre)	Grind-size + freshness videos	$1 \times$ length	~8 min	~0.15 (pre-class)
Live	Welcome + one model + taste reference	1:1	15 min	0.25
Live	Gear & beans walkthrough	1:1	15 min	0.25
Hands-on	Brew-along demo	1 brew/0.25 hr	1	0.25
Hands-on	Paired fill-in brew	1 brew/0.3 hr	1	0.30
Hands-on	Two-cup compare + tasting	real time	20 min	0.35
			Live total	~1.4 hrs

Read-out: ~1.4 hrs of live activity against a 1.5-hr workshop block — fits, but tight once you add transitions. If it had hit 2 hrs, move the two-cup compare to post-class rather than cut it. Pre-class video (~0.15 hr) sits well under any reasonable pre-work cap.

(Want the live spreadsheet behavior? Recreate these columns in your own spreadsheet tool.)